LS107 - GENETICS

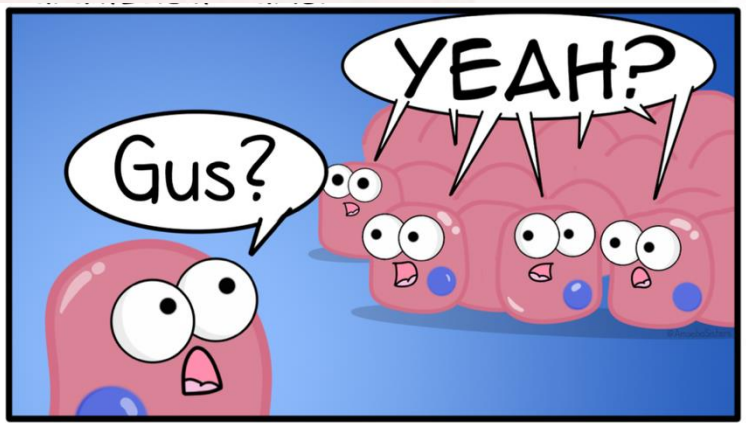
WEEK 4 DISCUSSION

Mitosis, Meiosis, and Chromosomal Nondisjunction

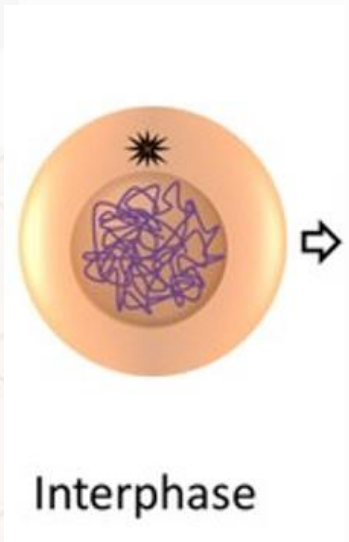
REMINDERS & UPDATES

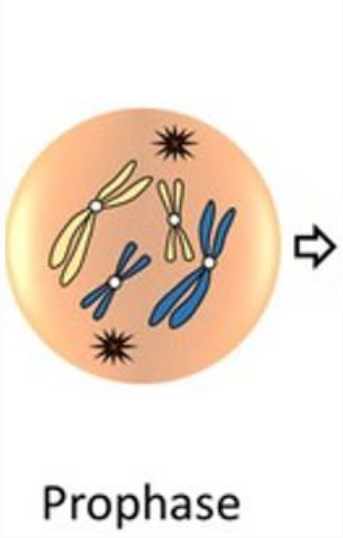
- Next Monday (01/31)
  - Midterm 1 (covers concepts from class and discussion worksheets, no linkage questions yay!!!
  - Weekly quiz (can be accessed during the weekend)
- Mode of instruction
  - Online for week 5
  - If you cannot make discussion please send me an email
- Worksheet 3 & Review

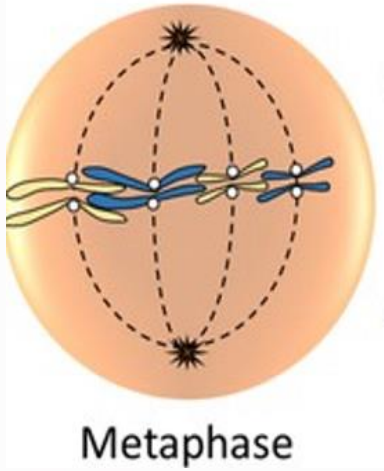
MITOSIS

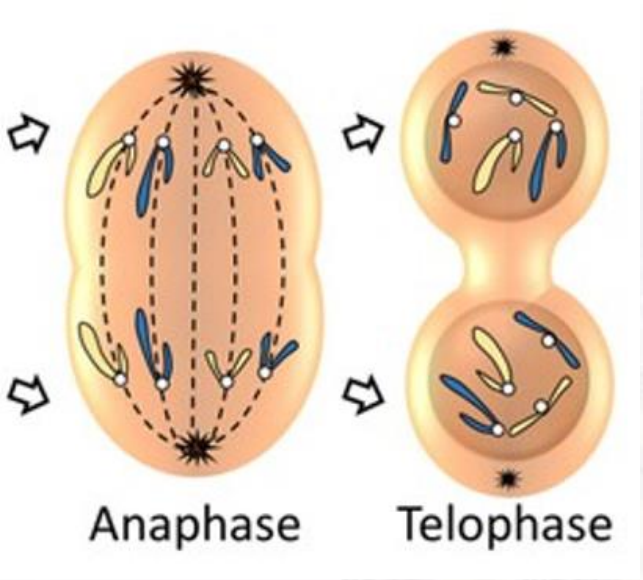


Mitosis: all fun and games until you lose your friend.


Interphase

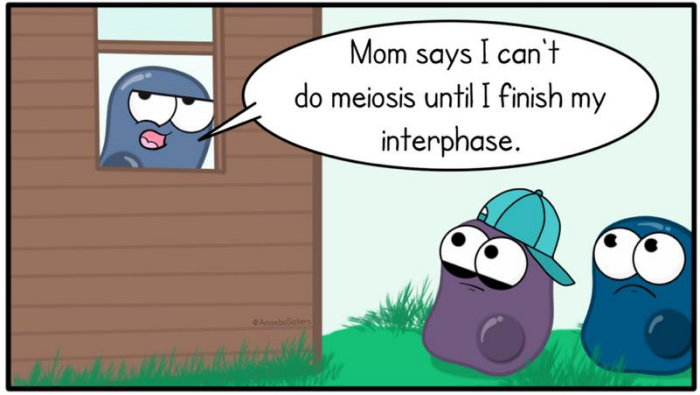

Prophase

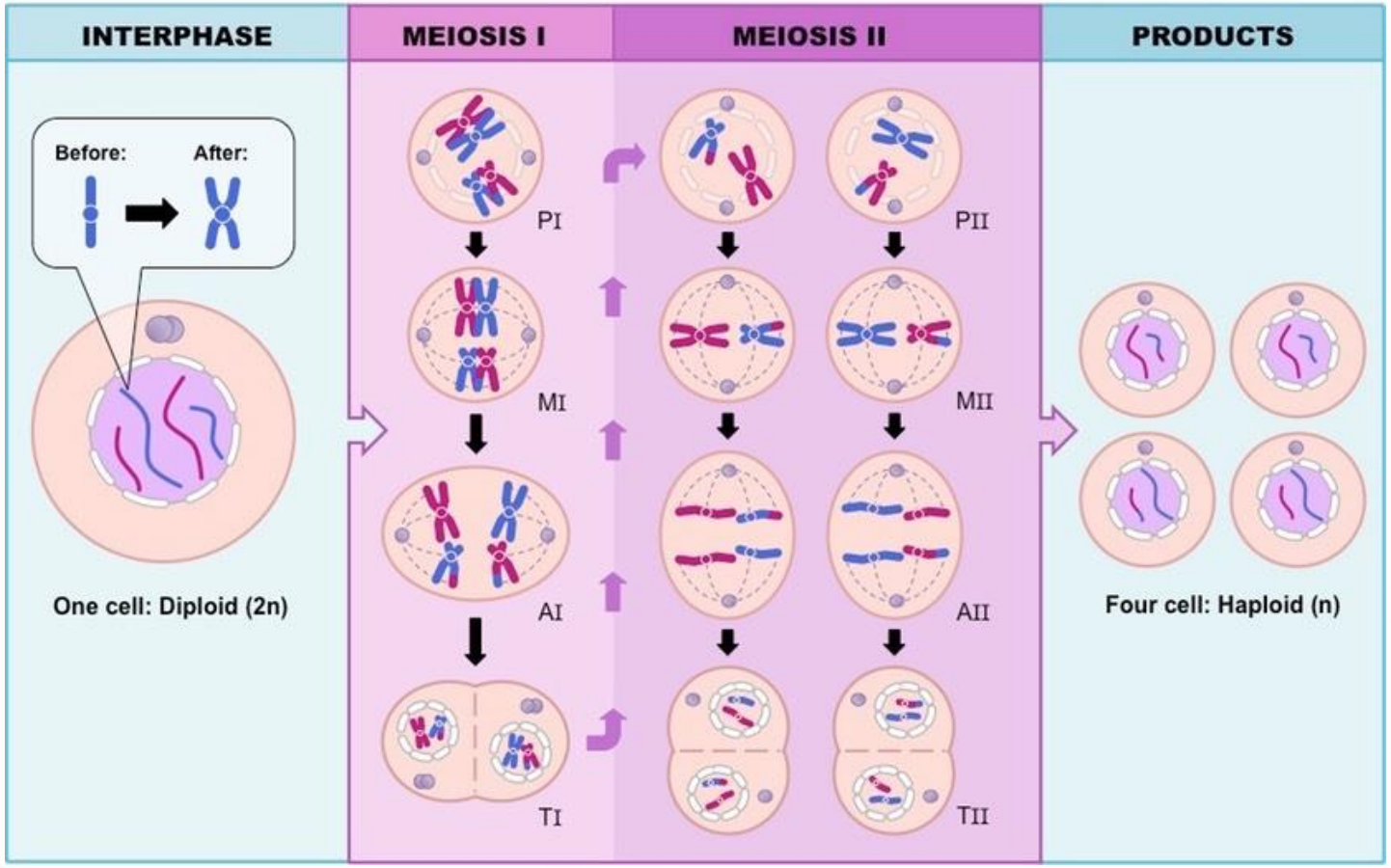

Metaphase


Anaphase

Telophase

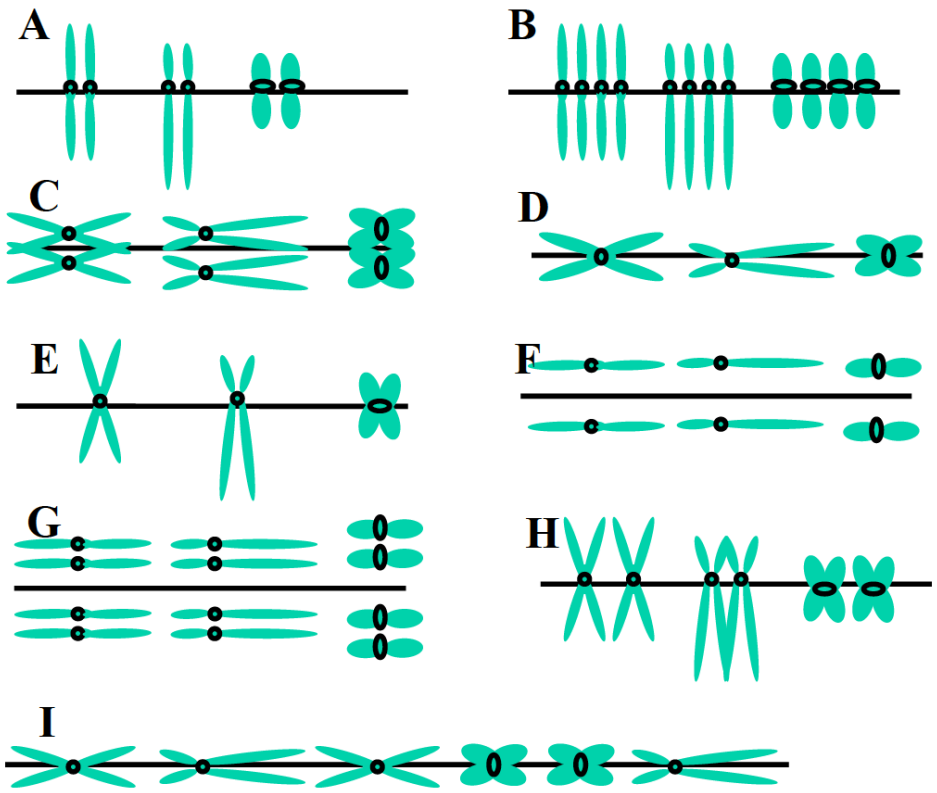
MEIOSIS


Mom says I can't do meiosis until I finish my interphase.


INTERPHASE
MEIOSIS I
MEIOSIS II
PRODUCTS
Before:
After:
One cell: Diploid (2n)
PI
MI
AI
TI
PII
MII
AII
TII
Four cell: Haploid (n)

WORKSHEET 3 – Q1

1. A diploid organism has three pairs of chromosomes (2n = 6). The lines in the figures below represent the metaphase plate that is perpendicular to this page. The circles represent centromeres.

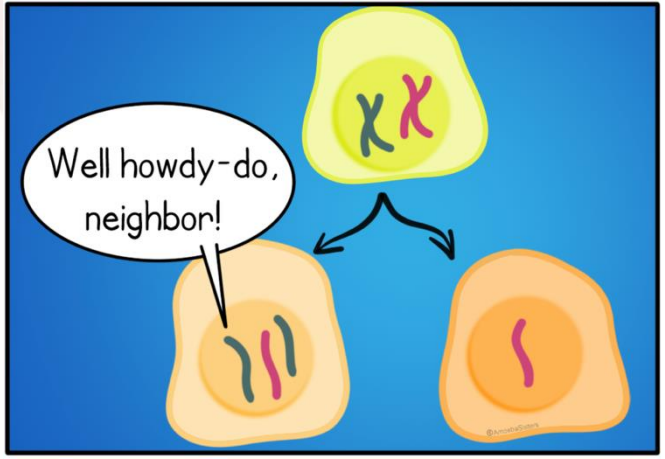


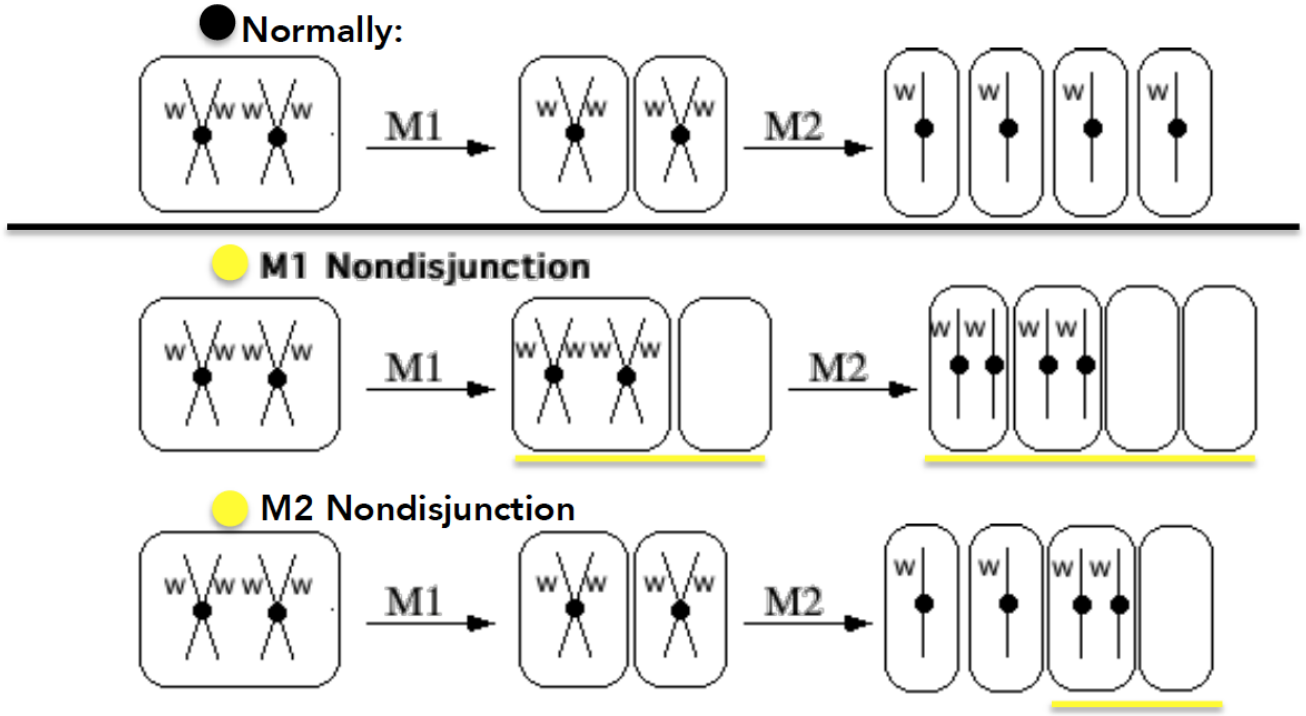
A. Which one of the above represents Metaphase during Meiosis I? _____

B. Which one of the above represents Metaphase during Mitosis? _____

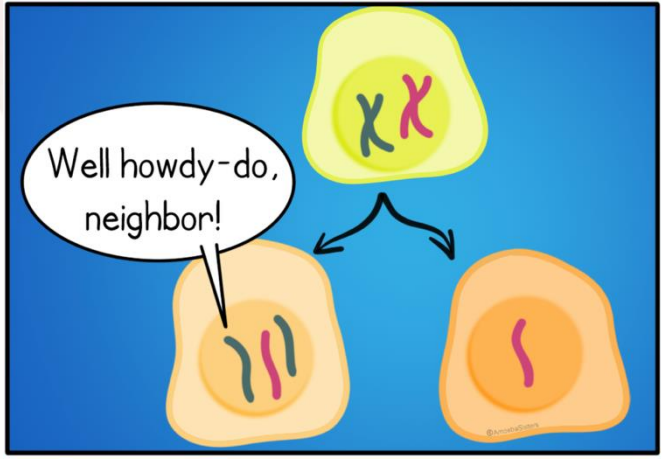
C. Which one of the above represents Metaphase during Meiosis II? _____

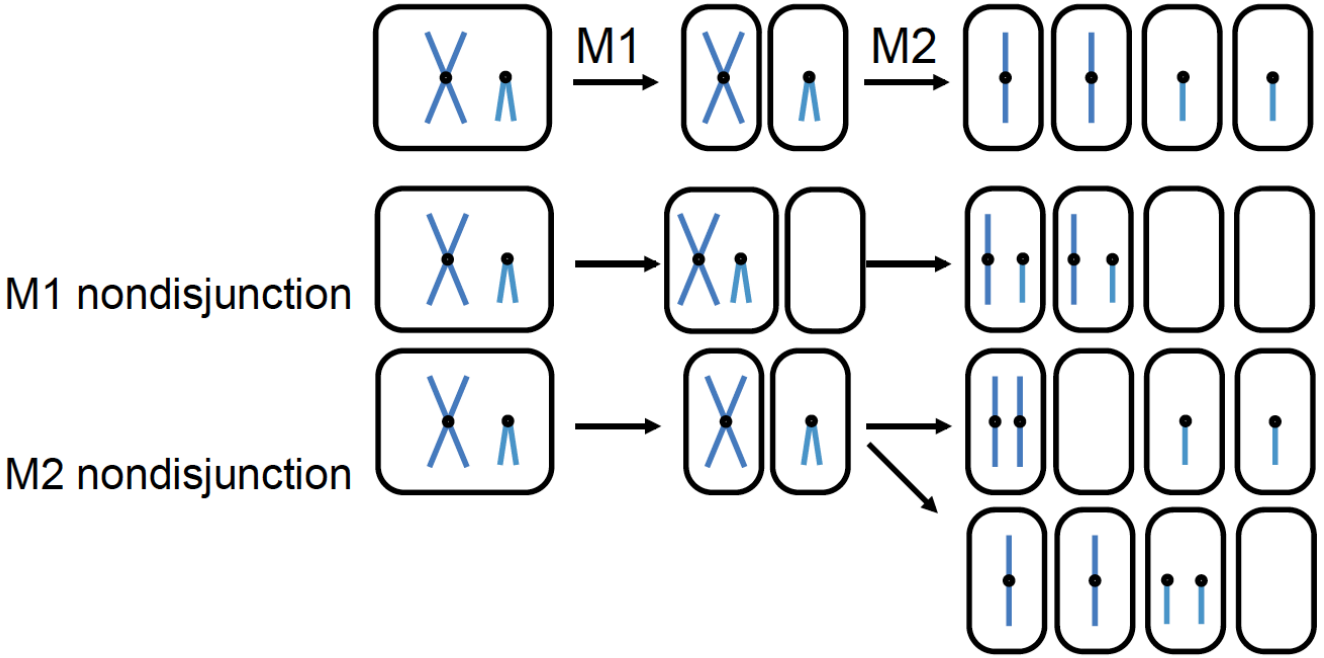
NON – DISJUNCTIONS





NON – DISJUNCTIONS





WORKSHEET 3 – Q2

2. The gene for nose shape is found on the X chromosome. Round nose is dominant to pointed nose.

Human individuals with XXY (an additional X chromosome) are male. Individuals with XO (only one X chromosome) are female. Individuals with XXX (three X chromosomes) are female.

For each of the following families, identify the possible nondisjunction events (**VERY RARE** mistakes during meiosis) that could explain the phenotype of the child. Circle one answer for each parent. *(Select "neither" if nondisjunction in that parent could not produce the child's phenotypic outcome. For example, if you think nondisjunction only in the father could explain the child's phenotype, then circle "neither" for the mother).*

WORKSHEET 3 – Q2

a) A man with a round nose and a woman with a pointed nose have a daughter with a round nose. This daughter has Turner Syndrome (genotype XO; one X chromosome).

Father (circle one): Meiosis I Meiosis II Either I or II Neither

Mother (circle one): Meiosis I Meiosis II Either I or II Neither

WORKSHEET 3 – Q2

b) A man with a round nose and a woman with a round nose have a son with Klinefelter Syndrome (genotype XXY) with a pointed nose.

Father (circle one): Meiosis I Meiosis II Either I or II Neither

Mother (circle one): Meiosis I Meiosis II Either I or II Neither

WORKSHEET 3 – Q2

c) A man with a round nose and a woman with a pointed nose have a son with Klinefelter Syndrome (genotype XXY) with a round nose.

Father (circle one): Meiosis I Meiosis II Either I or II Neither

Mother (circle one): Meiosis I Meiosis II Either I or II Neither

WORKSHEET 3 – Q2

d) A man with a pointed nose and a woman with a round nose have a daughter with a pointed nose. The genotype of this daughter is XXX (metafemale).

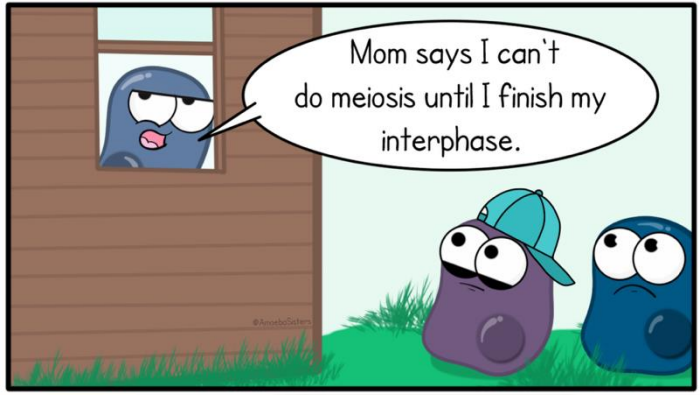
Father (circle one): Meiosis I Meiosis II Either I or II Neither

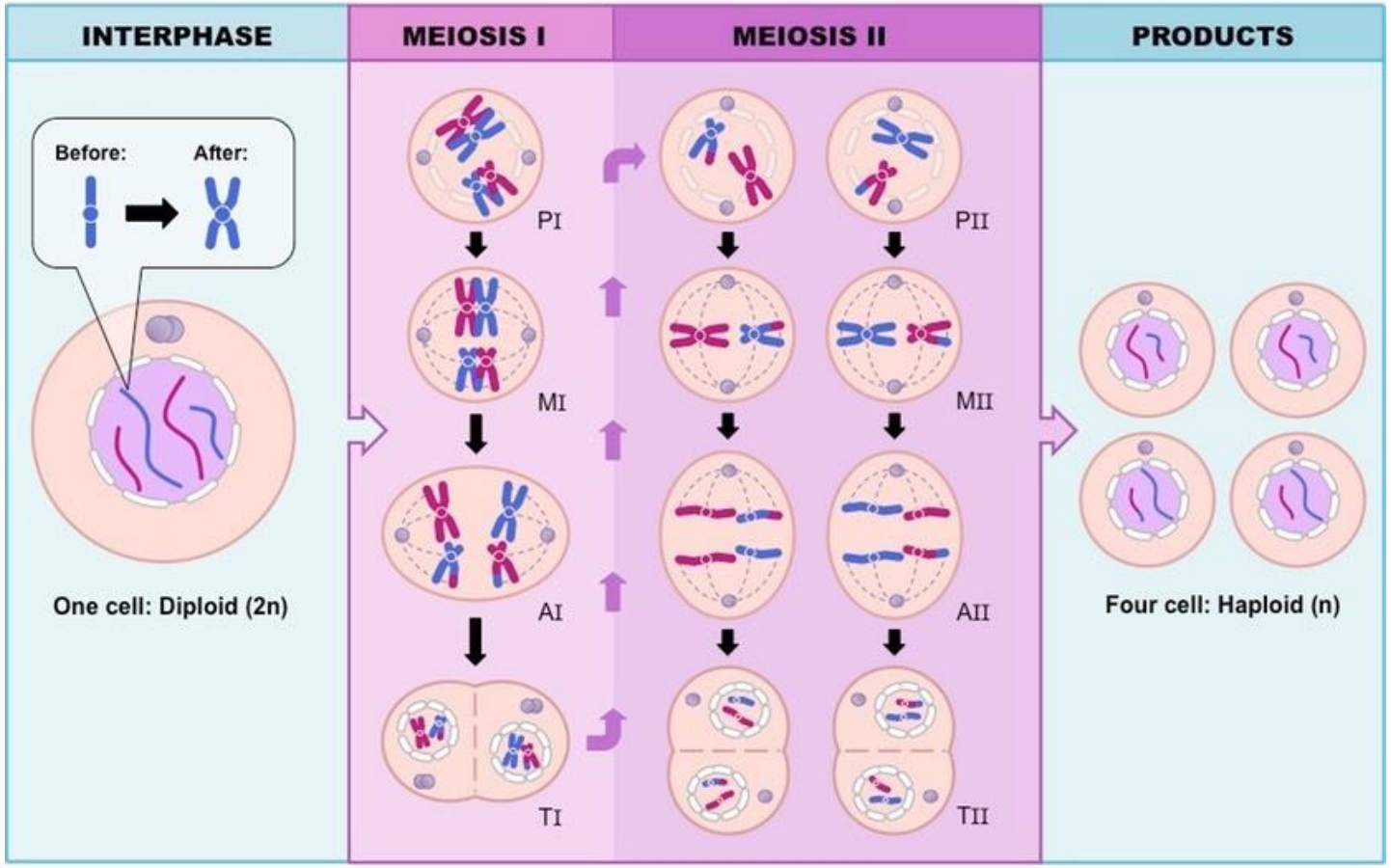
Mother (circle one): Meiosis I Meiosis II Either I or II Neither

WORKSHEET 3 – Q3

3. An organism has alleles *A* and *a* on one pair of homologous chromosomes, and it has alleles *B* and *b* on another pair. Diagram these pairs of homologs at the end of anaphase I. If there are multiple scenarios, just draw one.

MEIOSIS

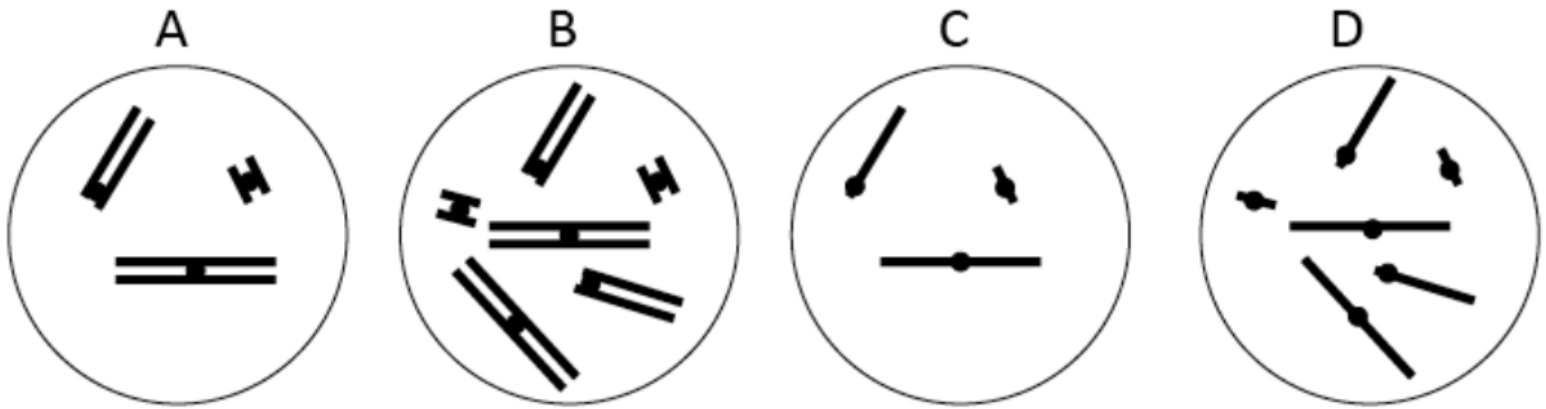

Mom says I can't do meiosis until I finish my interphase.


INTERPHASE
MEIOSIS I
MEIOSIS II
PRODUCTS
Before:
After:
One cell: Diploid (2n)
PI
MI
AI
TI
PII
MII
AII
TII
Four cell: Haploid (n)

WORKSHEET 3 – Q4

4. A. Which of the below shows the chromosomes in a cell that has just completed meiosis I?
(Circle one)

A **B** **C** **D** **None of them**



B. How many chromosomes does that cell have?_____

WORKSHEET 3 – Q5

5. Guppy has a diploid number of 2n = 46 chromosomes. For each of the following stages, identify the number of chromosomes present in each cell.

a. end of mitotic metaphase:_____

b. meiotic metaphase II:_____

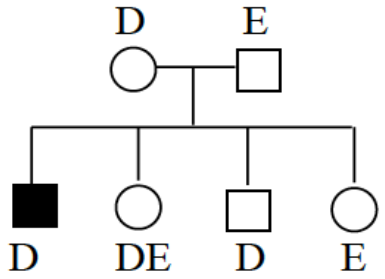
c. end of meiotic anaphase I: _____

d. end of mitotic anaphase: _____

WORKSHEET 3 – Q6

6. Consider the following pedigrees from human families containing a male with Klinefelter syndrome (a set of abnormalities seen in XXY individuals; indicated as black boxes). In each, 1^D and 1^E refer to codominant alleles of an X-linked gene. 1^F is recessive to both alleles 1^D and 1^E. The **phenotypes** of each individual (D, E, F or DE) are shown in the pedigree. Indicate if non-disjunction occurred in the mother or the father of the child affected with Klinefelter syndrome for each of the three examples. Can you tell if the non-disjunction was in the first or the second meiotic division? Fill answers into the provided boxes.

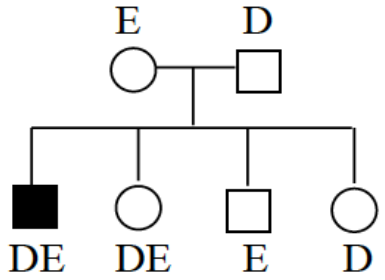
A.



nondisjunction in mother or father, or can be in either

non-disjunction in meiosis I or II, or can be in either

B.



nondisjunction in mother or father, or can be in either

non-disjunction in meiosis I or II, or can be in either

ATTENDANCE IN CHAT

WORD OF THE DAY...

GARFIELD!!!



